Consider the reaction between cerium metal and hot water shown below.

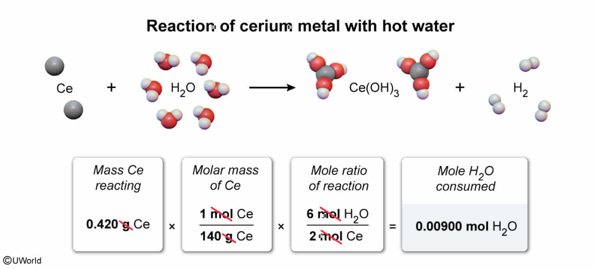
$$2\ \mathrm{Ce}(s) + 6\ \mathrm{H_2O}(l) \rightarrow 2\ \mathrm{Ce(OH)_3}(s) + 3\ \mathrm{H_2}(g)$$

How many moles of water are consumed in the reaction if 0.420 g of cerium fully reacts?

- A. 1.00×10^{-3} mol [13%]
- B. 3.00×10^{-3} mol [11%]
- C. 4.50×10^{-3} mol [8%]
- D. 9.00×10^{-3} mol [67%]

Correct
66% answered correctly

Explanation:



The molar mass of a compound is the amount of **mass contained in 1 mole** of the compound (grams per mole) and can be used to convert the mass of a compound to the number of moles. The number of moles of a chemical species is related to the number of moles of another chemical species in a reaction by the mole ratios of the stoichiometric coefficients from a balanced reaction equation.

For the reaction in this question, if 0.420 g of Ce(s) reacts, the number of moles of Ce(s) participating in the reaction can be calculated by dividing the mass of Ce(s) by its molar mass from the periodic table.

$$0.420\ \cancel{\mathrm{g\ Ce}(s)} \times \frac{1\ \mathrm{mol\ Ce}(s)}{140\ \cancel{\mathrm{g\ Ce}(s)}} = 0.00300\ \mathrm{mol\ Ce}(s)$$

The given balanced reaction equation shows that 6 mol of H_2O(l) are consumed for every 2 mol of Ce(s) that react. Applying this mole ratio, the number of moles of H_2O(l) consumed in the reaction is:

$$0.00300\ \cancel{\mathrm{mol\ Ce}(s)} \times \frac{6\ \mathrm{mol\ H_2O}(l)}{2\ \cancel{\mathrm{mol\ Ce}(s)}} = 0.00900\ \mathrm{mol\ H_2O}(l) \quad \Longrightarrow \quad 9.00 \times 10^{-3}\ \mathrm{mol}$$

(Choice A) A quantity of 1.00×10^{-3} mol is obtained if the mole ratio $\frac{6 \text{ mol } H_2O(l)}{2 \text{ mol Ce}(s)}$ is mistakenly inverted as $\frac{2 \text{ mol Ce}(s)}{6 \text{ mol } H_2O(l)}$ during the second step of the calculation.

(Choice B) A quantity of 3.00×10^{-3} mol is the number of moles of *cerium* consumed in the reaction.

(Choice C) A quantity of 4.50×10^{-3} mol is the number of moles of *hydrogen gas produced* in the reaction.

Educational objective:
The molar mass (the amount of mass in 1 mole of a compound) permits conversions between a given amount of mass and the number of moles that it contains. Stoichiometric mole ratios from a balanced reaction equation relate the number of moles of a chemical species to the moles of another chemical species in a reaction.

Time spent: 0 sec	**QID**:403891
Subject: General Chemistry	**System:** Interactions of Chemical Substances